DELTA-SIGMA ASIC DEVELOPMENT

Team_76

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I. Introduction

With the increasing reliance on digital processing in modern electronic systems, accurate conversion of real-world analog signals into the digital domain has become a necessity. Sensors used in different applications, such as environmental monitoring, biomedical instrumentation, industrial automation, and space systems, typically generate analog signals that must be digitized before storage, processing, and transmission. Thus, the accuracy, performance, noise suppression, and dependability of the analog-to-digital converter (ADC) used at the interface are critical for these systems' performance.

In many such applications, the signals of interest are of small bandwidth but demand very high precision rather than sampling extremely fast signals, the aim is to distinguish very small amplitude variations in the presence of noise, that is, to get an accurate, high-resolution output. For these systems, metrics such as signal-to-noise ratio (SNR), signal-to-noise and distortion ratio (SNDR) and effective number of bits (ENOB) are far more important than raw sampling speed. Achieving high ENOB, however, becomes increasingly challenging as system requirements scale, due to limitations imposed by quantization noise and circuit non-idealities.

Unlike SAR (Successive Approximation Register) ADCs which determine each bit through a fast binary search process and are well suited for moderate to high speeds, Delta–Sigma ($\Delta\Sigma$) converters take a fundamentally different route by shifting most of the quantization noise outside the signal band by operating the modulator at a sampling rate much higher than the signal bandwidth. A low-resolution quantizer is used inside a feedback loop, and the high-rate output bitstream is later processed by a digital decimation filter to recover a high-resolution digital signal. This approach makes Delta–Sigma ADCs particularly suitable for applications where high resolution and good SNDR are required at relatively low output data rates.

A key characteristic of Delta–Sigma ADCs is the inherent trade-off between sampling rate and effective resolution. Higher ENOB is typically achieved by increasing the oversampling ratio, which improves noise shaping and reduces in-band quantization noise. However, this often comes at the cost of operating at lower effective Nyquist rates or requiring aggressive decimation. Conversely, as the sampling rate is increased to support higher data rates, the achievable ENOB can degrade due to reduced noise-shaping benefit, increased sensitivity to non-ideal effects, and limitations in the digital filtering stage. A clear tension exists between achieving high sampling rates and maintaining high ENOB (which is essentially a function of SNDR). [1]

$$ENOB = \frac{SNDR_{dB} - 1.76}{6.02} \tag{1}$$

The motivation of this project arises directly from this trade-off. While conventional Delta–Sigma ADC designs typically achieve very high ENOB at low sampling rates, many practical systems require higher sampling rates without sacrificing resolution. The central problem addressed in this work is, therefore, to study and improve the ENOB achievable at higher sampling rates.

This project aims to thoroughly explore and select the most suitable Delta–Sigma modulator structures and digital filter designs. Different combinations of modulators and filters will be evaluated to see how they affect key performance metrics like ENOB, especially at higher sampling rates. The project will also implement the chosen digital or decimation filter using HDL, taking it through the full ASIC design flow including synthesis, layout, and post-layout simulation. Additionally, A digital filter model in MATLAB/Simulink will be developed and translated into RTL using Verilog or VHDL, ensuring complete system-level validation within the Delta–Sigma ADC framework.

II. Architecture & Methodology

The complete implementation flow from system-level modeling to ASIC layout is illustrated in Fig. 1. A third-order, 1-bit hybrid continuous-time/discrete-time $\Delta\Sigma$ modulator operating at a sampling frequency of 4.096 MHz constitutes the analog front-end. The loop filter employs a mixed CIFF - CIFB topology, with the noise-transfer function designed for an out-of-band gain of 1.5 and the signal-transfer function constrained to 0 dB DC gain. Behavioral modeling explicitly accounts for thermal noise, flicker noise, and DC offset to capture dominant circuit-level non-idealities. The resulting 1-bit DPCM stream is processed by a three-stage digital decimation chain consisting of a CIC decimator, a polyphase FIR compensation filter, and a half-band filter to achieve noise-shaping suppression, passband equalization, and final rate reduction. The digital architecture supports dual operating modes to trade resolution for bandwidth under a fixed sampling clock. The synthesizable RTL implementation is verified through functional simulation and subsequently mapped to a standard-cell library with full timing closure under the target clock constraint. The post-synthesis netlist is carried through physical design, including placement and routing, yielding a congestion-free layout that satisfies the prescribed area and timing budgets. The final ASIC implementation preserves the system-level performance targets in terms of OSR, Nyquist bandwidth, and effective resolution.

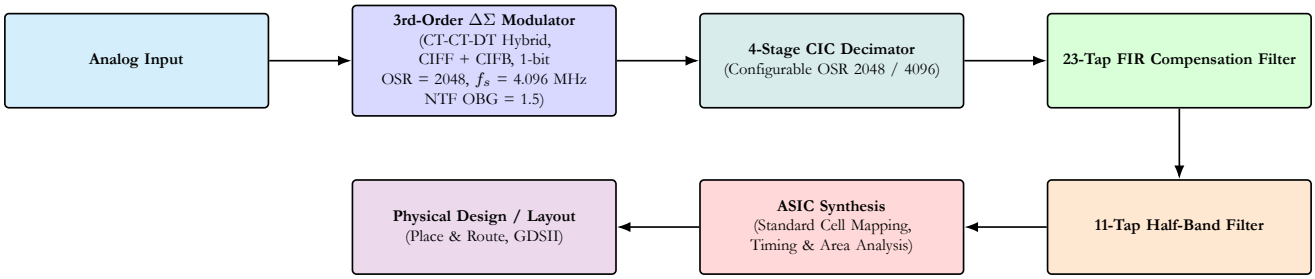


Figure 1: Overall design and implementation flow.

III. Modulator Study

The following section outlines the systematic design steps and theoretical studies conducted for the modulator implementation.

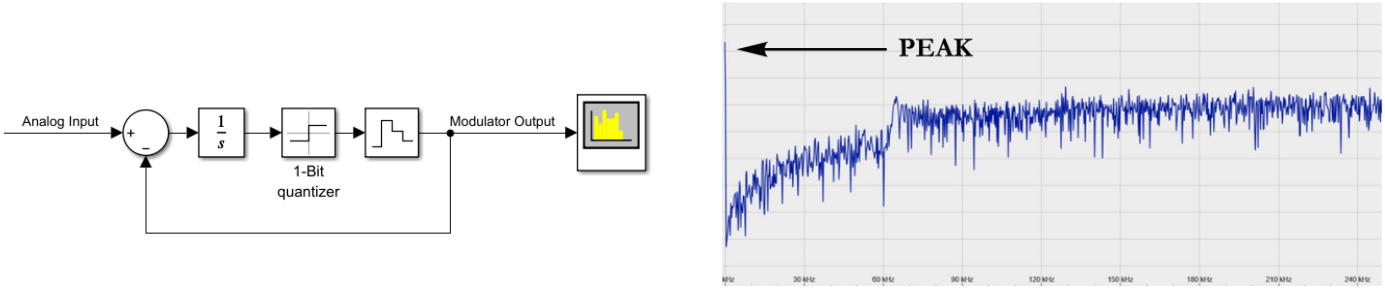
The design process commenced with the simulation of a conventional first-order continuous-time $\Delta\Sigma$ modulator to verify its fundamental operation and the expected noise-shaping characteristics.

As illustrated in Fig. 5b, the simulation successfully confirmed the noise shaping of quantization noise. The noise power is effectively suppressed in the low-frequency signal band (near the 1 kHz tone) and pushed towards higher frequencies.

Synthesis of Noise Transfer Function (NTF) and Order Selection of modulator

To determine the optimal modulator order, the Delta-Sigma Toolbox of MATLAB [2] was utilized for NTF synthesis and SNDR estimation. The Noise Transfer Function was synthesized targeting a maximum Out-of-Band Gain (OBG) of 1.5 to ensure stability. Subsequent simulations conducted at an Oversampling Ratio (OSR) of 256 demonstrated that a third-order modulator architecture is sufficient to achieve an Effective Number of Bits (ENOB) exceeding 16. The results of SNDR simulation of the synthesised NTF is shown in fig.3. The NTF synthesised for third-order modulator is given by

$$\mathrm{NTF}(z) = \frac{(z-1)^3}{(z-0.6694)(z^2-1.531z+0.6639)} \tag{2}$$



(a) Simulink model of the First-Order CT $\Delta\Sigma$ Modulator

(b) Observed Noise Shaping with 1 kHz Input

Figure 2: Preliminary verification of First-Order Noise Shaping

Study of different modulator topologies (CIFB and CIFF)

The circuit shown in fig.4a contains cascaded integrators with distributed feedback (CIFB), as it contains a cascade of N delaying integrators, with the feedback signal being fed to each integrator input terminal with different gains. In the CIFF topology fig.4b, each integrator's output is passed directly to a point where all signals are combined, just before the quantizer. Unlike the CIFB (Feedback) topology, where the DAC feedback connects to every integrator, the CIFF(Feedforward) approach typically connects the DAC feedback only to the input of the first integrator. [3] A comparison between both topologies is given in Table 1. [4]

Types Explored: CT, DT, HYBRID

In a discrete-time delta-sigma ADC, sampling of the input signal x(t) occurs prior to the loop filter . The loop filter H(z) is discrete-time, implemented using switched capacitor integrators. The loop filter shapes the quantization noise out of the baseband to higher

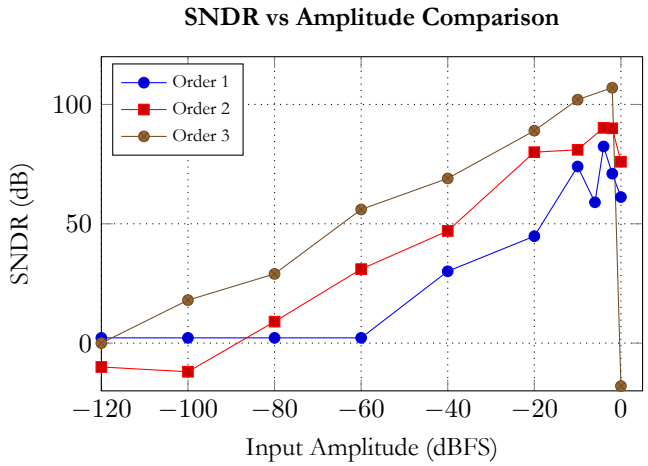
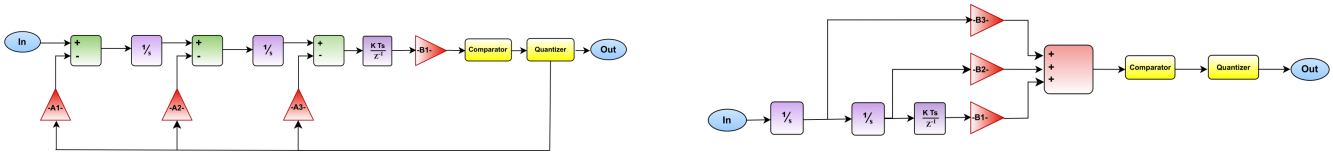


Figure 3: SNDR simulation result of synthesised NTFs of order 1, 2, and 3

Table 1: Comparison of CIFB and CIFF Modulator Topologies

Parameter	CIFB	CIFF
Internal Swing	High	Low
Power	High	Low
STF	Flat	Peaking
Complexity	Lesser than CIFF	More due to Extra Adder
Anti-Aliasing	Good	Poor



(a) Cascade of Integrators with Distributed Feedback (CIFB).

(b) Cascade of Integrators with Feedforward (CIFF).

Figure 4: The two Modulator Topologies

frequencies [5]. In a continuous-time delta-sigma ADC, noise shaping and oversampling work just like they do in the discrete-time version. The main difference is when the sampling happens. In a continuous-time design, the input is sampled right before the quantizer, while for the discrete-time design, it happens when the signal enters the first integrator. Table 2 summarizes the major differences between DT and CT modulators. [6]

Feature	Discrete-Time (DT)	Continuous-Time (CT)
Aliasing	**Poor**: Input sampled immediately. Needs external Anti-Aliasing Filter.	**Inherent**: Loop filter acts as an AAF, suppressing blockers before sampling.
Clock Jitter	**Robust**: Performance is generally insensitive to clock edges.	**Sensitive**: DAC waveform errors due to jitter directly degrade SNDR.
Tolerance	**Precise**: Relies on accurate capacitor matching. No tuning needed.	**Variable**: RC constants vary (±20%). Requires tuning/calibration.

Table 2: Major Differences Between Discrete-Time (DT) and Continuous-Time (CT) Modulators

Hybrid Modulator Architecture

The hybrid Delta-Sigma modulator represents a strategic architectural synthesis, effectively bridging the gap between Continuous-Time (CT) and Discrete-Time (DT) implementations. By employing a CT integrator for the critical input stage and shifting to a DT Switched-Capacitor (SC) topology for the remaining backend, this design optimizes the trade-offs inherent in both domains.

The specific advantages of this mixed-domain approach include:

- **Inherent Anti-Aliasing:** Leveraging a continuous-time first stage imparts inherent anti-aliasing characteristics to the ADC. This intrinsic filtering capability significantly relaxes, or even eliminates, the requirement for a power hungry, high performance external anti aliasing filter, addressing a primary limitation of pure DT architectures.
- **Simplified Input Drive:** The resistive input of the CT first stage presents a benign load compared to the switched capacitor inputs of pure DT designs. This eliminates the harsh switching transients and kickback noise associated with sampling, making the ADC significantly easier to drive without requiring high-power buffer amplifiers.

- **Containment of Jitter Sensitivity:** While the Continuous-Time input stage remains sensitive to clock jitter, the hybrid architecture confines this sensitivity to the CT stage only. The subsequent Discrete-Time backend is naturally robust against timing errors, making the overall system strictly less sensitive to clock jitter than a fully Continuous-Time loop.

- **Robustness to Process Variations:** By implementing the backend stages in the Discrete-Time domain, the design benefits from the superior matching properties of CMOS capacitors. This reduces the reliance on the precise absolute values of RC time constants, thereby minimizing the need for the complex calibration and tuning circuitry often required in high-order, pure CT designs.

Final topology chosen and justification

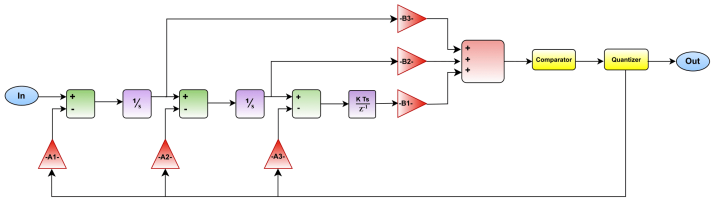
CIFF and CIFB architectures were combined to design the modulator as shown in Fig. 5a. The NTF of the modulator was derived and is given by Eq. 3 where the coefficients can be referred from Fig. 5a.

$$NTF(z) = \frac{(z-1)^3}{D(z)} \tag{3}$$

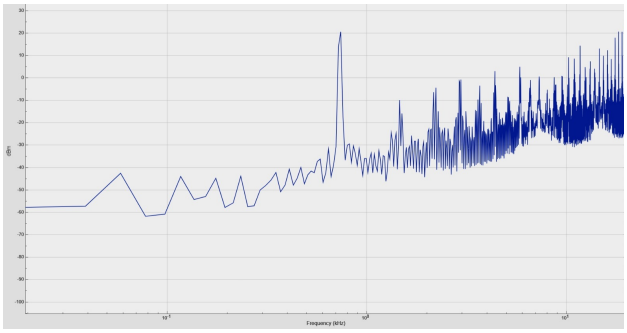
where

$$D(z) = z^3 - (3 + A_1B_3 - A_2B_2 - A_3B_1)z^2 + (3 - A_1B_2 + A_2B_1 + 2A_1B_3 - 2A_2B_2 - 2A_3B_1)z + (1 + A_1B_2 - A_2B_1 - A_1B_1 - A_1B_3 + A_2B_2 + A_3B_1)$$

Equating this NTF to the one synthesized for a 3rd order modulator using the Delta Sigma toolbox, three equations for six coefficients are obtained. This can be solved by setting a DC unity gain constraint and choosing the values of B1 and B2 as one. In this way all the coefficients were calculated to required NTF.



(a) Block diagram of the implemented Third-Order Delta-Sigma Modulator.



(b) Noise-shaping / output spectrum of proposed modulator.

Figure 5

Based on the comparative analysis of noise-shaping techniques, the optimal configuration was determined to be a hybrid CT-CT-DT cascade. This design integrates a mixed CIFF-CIFB loop filter to leverage the low-power benefits of feedforward paths while maintaining the pole-placement flexibility of feedback structures. The selection of a Third-Order Hybrid CIFF-CIFB Modulator is driven by the need to balance high-resolution performance with power efficiency and implementation robustness. Figure 5a shows the modulator design used. The specific technical justifications for this architectural synthesis are detailed below:

1. **Mitigation of Kickback Noise and Drive Requirements (Hybrid Advantage):** Unlike pure Discrete-Time (DT) architectures, where the input sampling capacitor creates significant switching transients (kickback noise), the proposed Continuous-Time (CT) first stage presents a purely resistive input impedance. This eliminates sampling glitches at the input node and significantly relaxes the settling and bandwidth requirements of the preceding sensor driver or buffer amplifier.

2. **Inherent Anti-Aliasing (CT Input Advantage):** The CT input stage functions as an inherent Anti-Aliasing Filter (AAF) prior to sampling. This effectively suppresses high-frequency blockers and noise, removing the system-level burden of implementing a steep, high-order external active filter, which is otherwise mandatory in pure DT implementations to prevent noise folding.

3. **Power Optimization via Signal Swing Reduction (CIFF Path):** The architecture incorporates feedforward (CIFF) paths, ensuring that the loop filter processes primarily the quantization noise rather than the full scale input signal. This results in significantly reduced output swings at the integrator nodes, allowing the Operational Transconductance Amplifiers (OTAs) to operate with relaxed slew-rate and linearity requirements, directly translating to reduced static power consumption.

4. **Stability and Matching Precision (DT Backend Advantage):** While the CT front-end handles the sensor interface, the backend integrators are implemented in the Discrete-Time (DT) domain. This leverages the superior matching properties of CMOS capacitor ratios (typically $< 0.1\%$) over the absolute RC time constants of CT circuits (which can vary by $\pm 20\%$). This hybrid approach eliminates the need for complex on-chip tuning circuitry for the higher order stages while maintaining the stability flexibility provided by the distributed feedback (CIFB) structure.

Tradeoff between Nyquist sampling frequency and ENOB

Simulation studies were undertaken to observe and confirm the tradeoff between the Nyquist sampling frequency and the Effective Number of Bits (ENOB). The modulator was driven by a sinusoidal test signal with an amplitude of 1 V and a frequency of 100 Hz. The resulting output bitstream was processed by a digital decimation filter, where the decimation ratio was systematically varied to sweep the effective Nyquist frequency. Table 3 details the specific results for a modulating frequency of 1.024 MHz, while Figure 6 illustrates the performance trade-offs across a broader range of modulating frequencies.

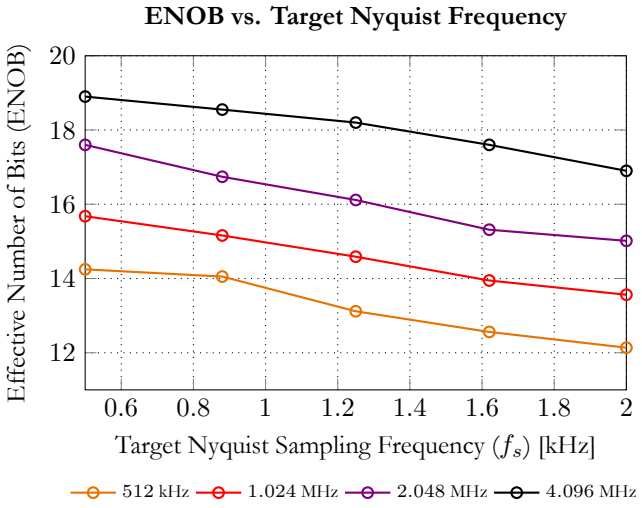


Figure 6: ENOB vs. Target Nyquist Frequency

OSR (r)	Actual f_s (kHz)	ENOB (bits)
8192	0.5000	18.900
4680	0.8752	18.550
3280	1.2488	18.200
2520	1.6254	17.600
2048	2.0000	16.900

Table 3: Simulation Results for $f_M = 4.096$ MHz

IV. Noise Analysis

Overview of Noise Sources

In low-frequency $\Delta\Sigma$ ADCs, the input OTA and first integrator are the primary noise sources. At target rates of 0.5–2 ksps, flicker noise dominates while thermal noise sets the floor. If we don't mitigate this noise, it will lower the effective resolution and put the system's stability at risk. The noise modeling is performed in MATLAB in order to capture these effects accurately. The model comprises:

- Base thermal noise (white) injected at OTA outputs.
- 1/f noise modelled using a power law PSD $S_v(f) = \dfrac{K_f}{f^\alpha}$ with time domain synthesis via inverse FFT to obtain the equivalent input referred flicker noise sequence.

Flicker Noise Impact on ENOB:

Simulink modeling shows that flicker noise dominates the 0.5–2 ksps baseband, causing an ENOB reduction of $\sim$ 0.25–0.46 bits (Fig. 8). The PSD exhibits a clear 1/f knee, confirming this noise source as the primary bottleneck. Consequently, mitigation techniques are mandatory to achieve the target resolution in this low-bandwidth context.

Noise modeling and DC Offset Analysis: We employed widely adopted analytical expressions [7] to model thermal, flicker, and offset noise, which is essential for accurate system-level evaluation of the $\Delta\Sigma$ modulator.

Thermal Noise modeling: Broadband thermal noise is modeled using the standard MOS channel approximation. The input referred voltage and current spectral densities are given by:

$$\overline{v_{n,\text{th}}^2}(f) = \frac{4kT\gamma}{g_m}, \qquad \overline{i_{n,\text{th}}^2}(f) = 4kT\gamma g_m. \tag{4}$$

This white-noise term establishes the high frequency noise floor for the behavioral modulator.

Flicker Noise modeling: Low frequency noise is captured using the unified $1/f$ formulation [7]:

$$\overline{v_{n,1/f}^2}(f) = \frac{K_f}{C_{ox}^2 WLf^\alpha}, \tag{5}$$

where $\alpha \approx 1$. This expression governs the inverse-FFT synthesis of the behavioral time series.

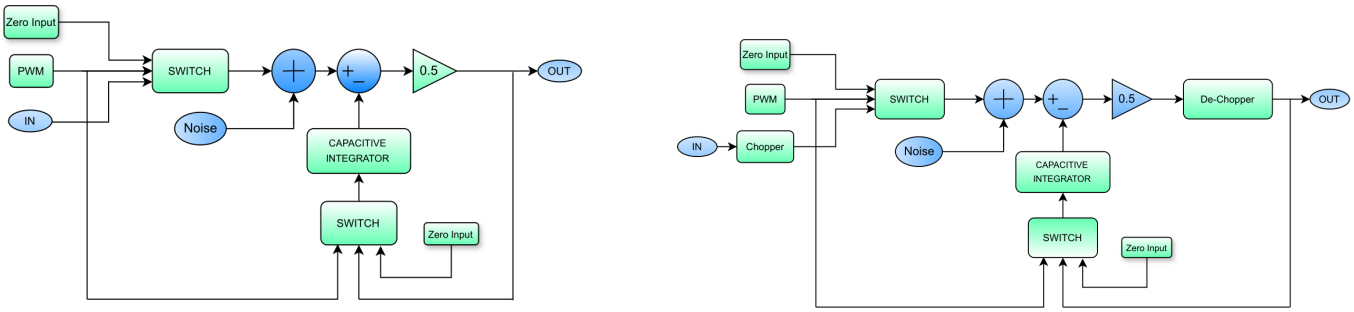
Combined Spectrum & Offset: The aggregate OTA input referred noise PSD is defined as the sum of the thermal and flicker components:

$$S_{v_n}(f) = \frac{4kT\gamma}{g_m} + \frac{K_f}{C_{ox}^2 WLf^\alpha}. \tag{6}$$

For DC offset, a fixed voltage is added to the signal path to model the worst case OTA input offset, enabling the evaluation of static error propagation.

Comparison of different noise mitigation techniques: All mitigation techniques were implemented using discrete-time switching primitives within Simulink:

Chopping: The behavioral modulation and demodulation blocks modulate the input such that flicker (1/f) noise and DC offset are translated to harmonics of the chopping frequency, thereby substantially suppressing the in-band 1/f component. As in [8], this frequency translation enables effective low-frequency noise removal, but the switching action and practical non-idealities introduce residual ripple and occasional folding of broadband noise. In our simulations, these effects limited the net benefit.



(a) Auto-Zeroing

(b) Chopping with AZ

Auto-Zeroing: A two-phase sample subtract structure ("auto-zeroing") removes most of the low-frequency (flicker and offset) noise, but as in [9] the sampling process inevitably folds higher-frequency white noise back into the baseband, raising the white-noise floor. As a consequence, when oversampling ratio (OSR) increases, the overall SNDR can degrade, and AZ may fail to yield a net ENOB improvement in our target bandwidth.

Nested Chopping: The residual noise artifacts that come with standard chopping were removed using a nested modulation architecture. This method eliminates residual baseband disturbances without causing the noise aliasing penalty of auto-zeroing by cascading high and low frequency modulation stages. Superior signal cleanliness with barely noticeable residual spurs is confirmed by spectral analysis.

Comparative Analysis of Noise-Reduction Techniques

Across Nyquist frequencies from 0.5 kHz to 2 kHz, five mitigation strategies were evaluated: Chopping (Chop), Auto-Zeroing (AZ), Nested Chopping (NC), Chopping+AZ, and NC+AZ.

- **Chopping (Chop):** Provided only modest ENOB gains ($\sim$ 0.2–0.38 bits), as performance remained limited by residual offsets and up-modulated spurs.
- **Auto-Zeroing (AZ):** Proved ineffective; the aliasing of broadband noise raised the overall noise floor, negating low-frequency suppression and degrading the net resolution.
- **Nested Chopping (NC):** This configuration yields an ENOB improvement of $\sim$ 0.2–0.45 bits . It resulted in an almost complete removal of residual noise artifacts in the PSD. Once these baseband components are suppressed below the thermal-noise floor, the in-band noise becomes predominantly white.
- **Chopping + Auto-Zeroing (Chop+AZ):** This hybrid method provides a moderate improvement of $\sim$ 0.2–0.4 bits at low f_s by combining up-modulation with offset cancellation. However, the sampling action associated with AZ introduces aliased broadband noise.
- **Nested Chopping + Auto-Zeroing (NC+AZ):** This combined method achieves ENOB gains comparable to nested chopping alone,$\sim$ 0.2–0.45 bit. While AZ further suppresses slow drift, behavioral simulations indicate that the overall in-band noise remains thermal-noise limited, resulting in no statistically significant ENOB advantage over NC.

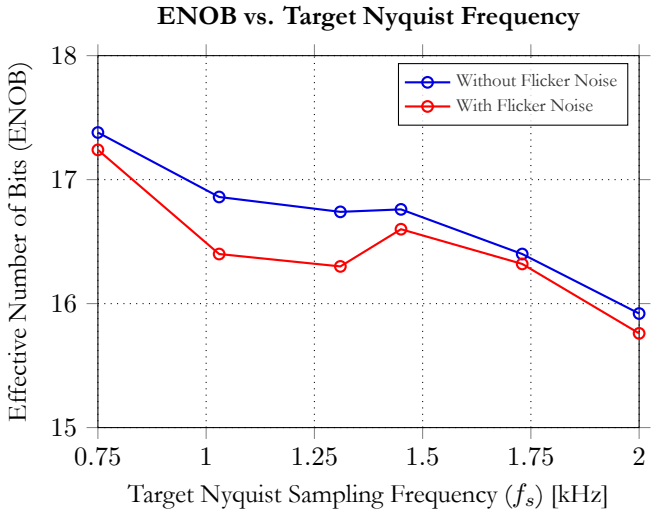


Figure 8: ENOB vs. Target Nyquist Frequency (with Flicker Noise)

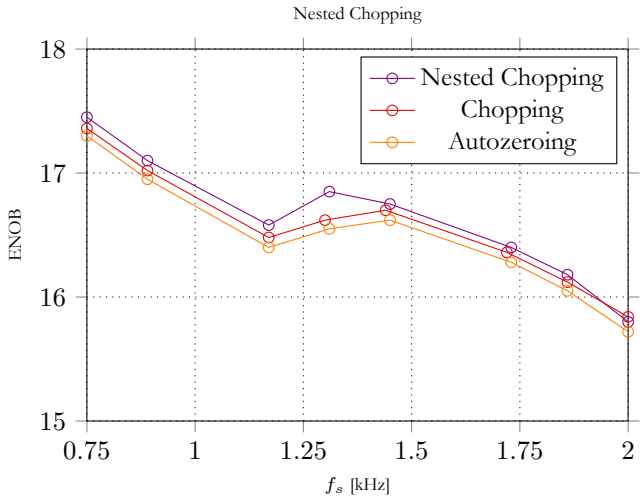


Figure 9: ENOB vs. target Nyquist frequency for chopping techniques

Final Selected Technique and Rationale

Nested Chopping (NC) was finalized as the front-end noise mitigation technique. By suppressing in-band noise well below the thermal floor. While the NC+AZ combination further reduces residual drift, it fails to improve ENOB as the system is already thermal noise limited. Thus, NC was chosen for its superior balance of robustness and implementation efficiency.

V. Digital / Decimation Filter

According to [10], achieving the required frequency characteristics in a single stage demands an extremely high filter order, leading to impractical hardware complexity (adders, multipliers, and registers). This issue is effectively mitigated by decomposing the decimation chain into a multi stage cascaded structure. The decimation filter implements a three-stage architecture. The filtering process is divided into three well distinguished stages: a Cascade Integrator-Comb (or SINC), a Compensation FIR filter, and a Half-band FIR filter. The decimation is primarily performed in the CIC filter stage; the FIR filter is used to compensate for the attenuation provided by the CIC filter in the passband. To further filter the noise, a Half-Band filter is used at the end. As a result, suitable architecture was created and outcomes were achieved, which is covered in the following sections.

Filter Architecture

Stage 1: Cascade Integrator-Comb (CIC) Filter: This includes a fourth-order CIC filter,fig.10 which performs the majority of the decimation and signal recovery. Since this architecture does not require any multipliers and use only adders and registers, it is highly capable of handling the high speed modulator output. However, the CIC magnitude response contains a droop in the passband, which needs to be compensated in subsequent stages.

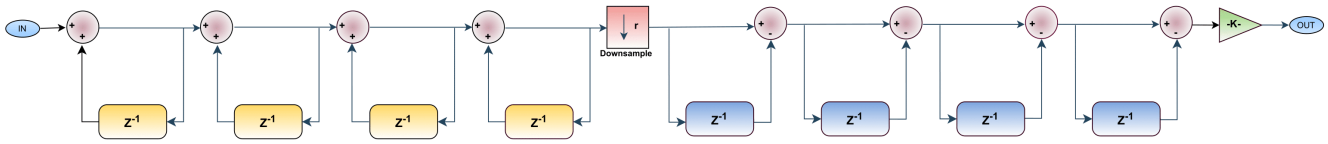


Figure 10: Block diagram of the Cascaded Integrator-Comb (CIC) Decimation Filter

Stage 2: Compensation FIR-Filter The second stage employs a 23-tap polyphase FIR-filter (Fig.11) specifically designed to equalize the passband droop inherent to the preceding CIC stage. While this architecture supports decimation, the proposed design configures the filter solely for spectral shaping without further downsampling. Its primary function is to sharpen the transition band and attenuate high-frequency quantization noise. The frequency response validating this design choice is illustrated in Figure 13.

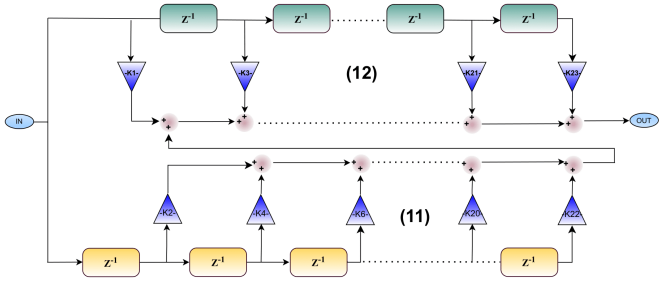


Figure 11: Block diagram of the FIR Filter

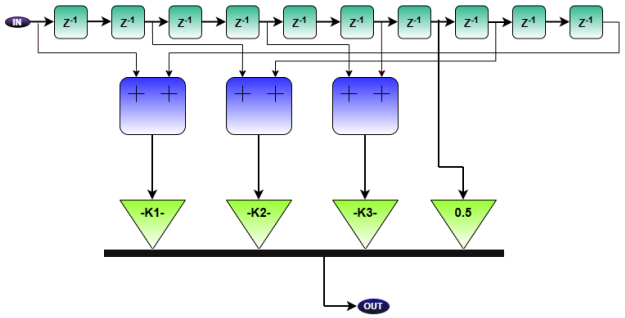


Figure 12: Block diagram of the FIR Half-Band Filter

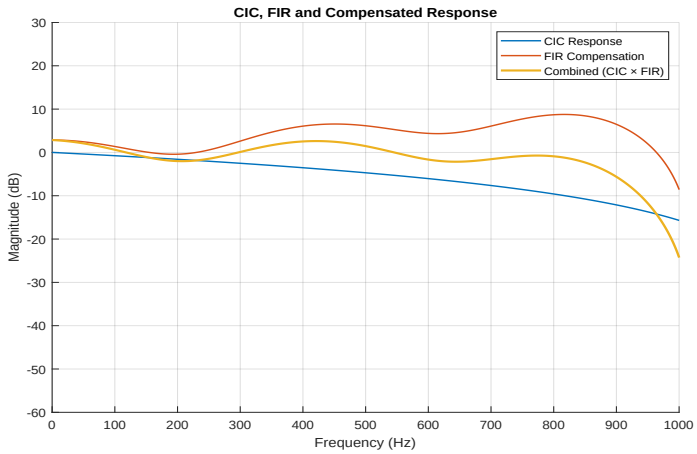


Figure 13: Frequency response analysis showing the uncompensated CIC droop (Blue), the inverse FIR compensation magnitude (Red), and the resulting flattened combined response (Yellow).

Stage 3: Half-band FIR Filter This is used in the last step for subsequent filtering of in-band noise. It provides good improvements in the final result as concluded from the simulation results. It is an extremely efficient architecture due to its characteristics, which reduce computing complexity (multiply-accumulate operations) by about 50%, as opposed to a standard FIR filter with similar specifications.

CIC Decimator

Higher order decimator filters are necessary to obtain the required downsampling for the FIR and smoothening of the modulator bitstream output. The transfer function of the filter is given as

$$H(z) = \left(\frac{1 - z^{-RM}}{1 - z^{-1}} \right)^N$$

In the above formula, R denotes the decimation factor, N denotes the order of the filter and M denotes the differential delay (which is usually 1). This means that the magnitude response will be composed of sinusoidal functions, given by:

$$|H(e^{j\omega})| = \left| \frac{\sin\left(\frac{RM\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} \right|^N$$

This filter outputs the signal with a significant gain, which is given by: [11]

$$G_{\max} = (RM)^N$$

Following the recommendations in [12], the CIC filter order must be at least one greater than the modulator order. Consequently, a fourth-order CIC filter was selected.

Coefficients of FIR Compensation filter

The coefficients for the FIR polyphase filter are obtained by solving for the fourier series of the FIR transform function, the solution of which is given by: [11]

$$h(n) = h(N - 1 - n) \tag{7}$$

This symmetry condition suggests that only half of the coefficients are unique and that the FIR has a linear phase response. The remaining coefficients are obtained by mirroring:

$$h(0) = h(N - 1),\ h(1) = h(N - 2),\ \ldots$$

From the multiple available windowing techniques tested (eg. Hann, Hamming, Kaiser) [13] [14] [15], the Parks-McClellan algorithm is chosen for deciding the coefficients of the FIR filter through MATLAB. The comparison of results for different numbers of taps (11, 23, 45, 91) for each FIR led to the conclusion that optimal performance was obtained at 23 taps, minimizing the truncation error in RLT design while achieving high accuracy in both.

The firpm library of MATLAB is used to find the tap values of the FIR filter [14]. The library comes from the Chebyshev approximation theory and the Remez exchange algorithm. The ultimate result of the entire algorithm is that a suitable equiripple FIR filter is obtained with passband and stopband ripples having equal amplitude.

Comparative analysis of FIR and IIR

Parameter	**IIR Filter**	**FIR Filter**
Number of multiplications	Fewer (due to recursion)	More (purely feed-forward)
Probability of overflow	High (feedback accumulation)	Very low
Stability	Can be unstable	Always stable
Phase response	Non-linear	Linear (with symmetric taps)
Hardware control complexity	Higher; recursion adds control overhead	Simple datapath and straightforward control
Adaptive filtering support	Achievable but requires careful stabilization	Excellent; widely used in adaptive algorithms

Table 4: Comparison of selected characteristics of IIR and FIR filters.

Considering these differences, FIR-based architectures were chosen due to their guaranteed stability, linear phase response, and suitability for multirate decimation chains in sigma–delta converters.Table 4 summarizes the key differences between IIR and FIR filters in the context of oversampled sigma–delta decimation chains.

Half-Band FIR Filter

As a specialized FIR topology, the half-band filter offers significant advantages over traditional designs, including a flat passband and excellent anti-aliasing properties. Its defining characteristic that every alternate coefficient is zero greatly simplifies the hardware implementation. This results in reduced logic utilization, lower power consumption, and a smaller footprint, making it the optimal choice for the final decimation stage [12].

Halfband filters also possess a highly efficient structure. Due to their inherent frequency symmetry, the impulse response satisfies [11]

$$h(n) = h(N-1-n),$$

and every alternate tap is exactly zero:

$$h(2k+1) = 0, \qquad \forall k.$$

As a result, the multiplier count is decreased by almost 50% and only roughly half of the coefficients are non-zero. Therefore, the halfband filter provides further anti-alias filtering while minimizing hardware complexity. Its combination of linear phase, structural efficiency, and ideal cutoff location makes it a natural and optimal choice in the final stage of the multirate decimation chain.

Final ENOB Evaluation of Simulink model at Optimal Modulating Frequency

After running comprehensive simulations, a final modulating frequency of 4.096 MHz was selected, Fig. 14, Fig. 15, and Fig. 16 show the results obtained at the output of the CIC Decimator filter, FIR Compensator, and HalfBand Filter for this modulating frequency, clearly marking the improvement in ENOB at each step.

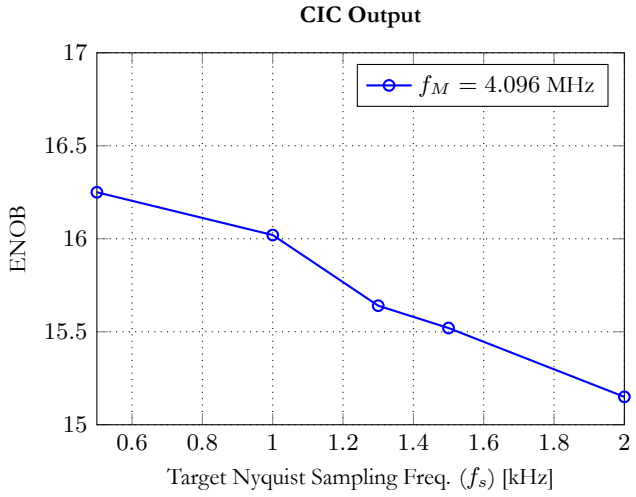


Figure 14: ENOB vs. Target Nyquist Frequency (CIC)

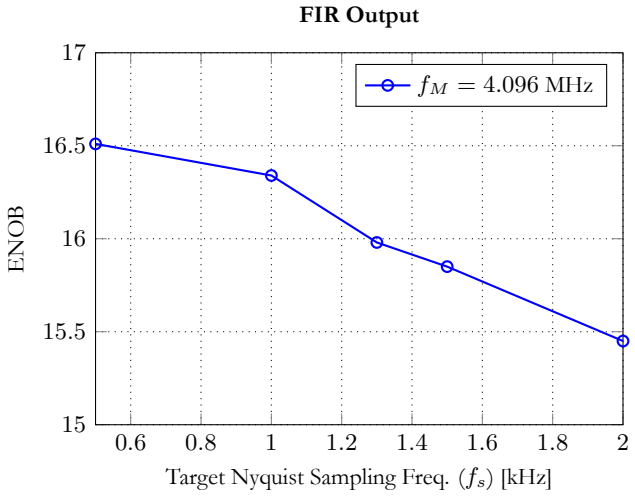


Figure 15: ENOB vs. Target Nyquist Frequency (FIR)

VI. HDL Design

RTL Architecture and Block Descriptions

The RTL design is created with synthesizable HDL and processes a 1-bit PCM signal from the Delta-Sigma modulator. This pcm data at the oversampling frequency is used as the input to the digital filtering chain, which is implemented in RTL through a series of decimation and filtering stages.

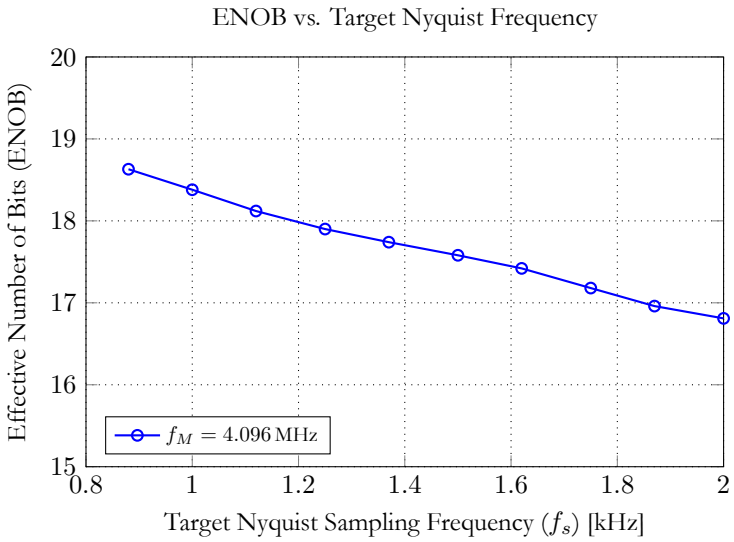


Figure 16: ENOB versus target Nyquist frequency for $f_M = 4.096\,\text{MHz}$ (final output).

The single-bit PCM data is first converted into a signed two's-complement format, where a binary '0' represents -1 and a binary '1' represents $+1$. This representation allows correct signed arithmetic in the filter stages. All delay elements required in the digital filters are implemented using clock-triggered D flip-flops, ensuring synchronous operation and suitability for synthesis and ASIC implementation.

The digital hardware supports two operating modes. Mode 1 corresponds to a high-resolution configuration with OSR = 4096. Mode 0 corresponds to a low-resolution configuration with OSR = 2048.

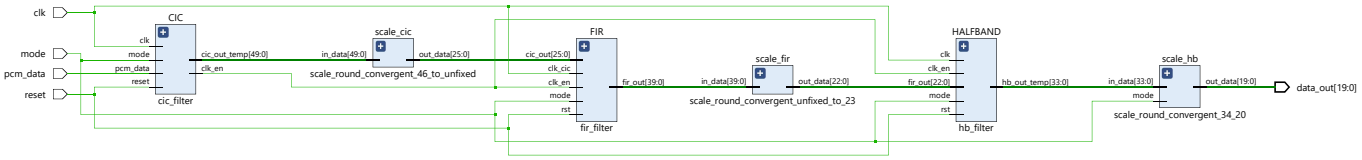


Figure 17: Block diagram of the Digital Filters Flow with Bits specification

The first stage of digital processing is a Cascaded Integrator-Comb (CIC) filter consisting of integrator stages operating at the oversampled clock, followed by a decimator with a factor equal to the oversampling ratio (OSR), and a corresponding set of differentiator stages operating at the reduced sampling rate. The internal word length of the CIC filter is determined analytically to avoid overflow in the integrator stages using [16].

$$W = 1 + InputBit + N \log_2(\text{OSR}) \tag{8}$$

where N denotes the number of CIC stages and one bit for Signed bit, in our case Input Bit is 1 bit PCM data and N=4. To support both operating modes (OSR = 2048 and OSR = 4096), the design is sized for the worst-case condition, resulting in a CIC word length of 50 bits to guarantee overflow-free operation up to the final integrator stage.

Since the large word length at the CIC output would significantly increase computational complexity in subsequent filters, truncation-based scaling is applied to reduce the word length from 50 bits to 26 bits before further processing. This scaled output serves as the input to the FIR filtering stage.

Following the CIC filter, a finite impulse response (FIR) filter is implemented to compensate for CIC passband droop and provide improved stopband attenuation. The FIR filter coefficients are designed in MATLAB and mapped to signed 11-bit integers using a common scaling factor to eliminate decimal arithmetic. The FIR filter in RTL is realized using a polyphase architecture, in which the arrangement of multipliers, adders, and delay elements is modified to perform filtering and decimation efficiently. Considering worst-case multiplication and accumulation, the internal FIR computation results in a maximum word length of approximately 40 bits. To limit growth and enable further processing, FIR output scaling using truncation is applied, reducing the word length to 23 bits.

The scaled FIR output is then applied to a half-band filter, which performs additional decimation and spectral refinement. Inter-

mediate computations in the half-band filter result in a maximum word length of 34 bits. Final scaling and truncation are applied at the output of the half-band filter, yielding a 20-bit digital output that represents the final ADC sample . A block diagram illustrating the entire digital processing chain along with word length transitions is shown in Fig. 17.

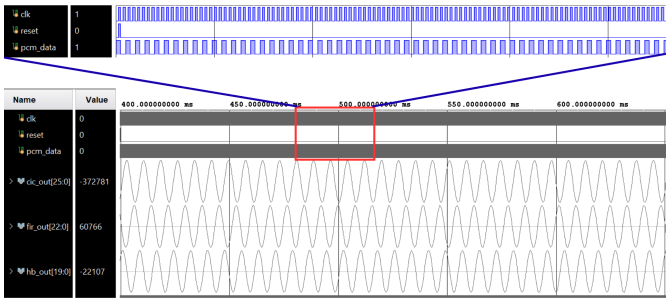


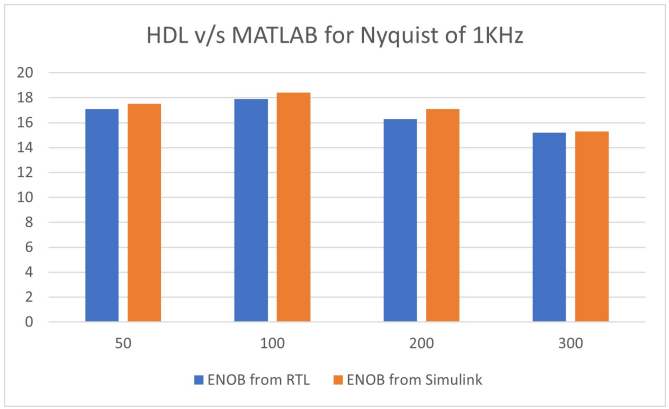
Figure 18: Simulation result at the CIC, FIR, and half-band outputs.

Table 5: RTL Performance for Different Nyquist Frequencies

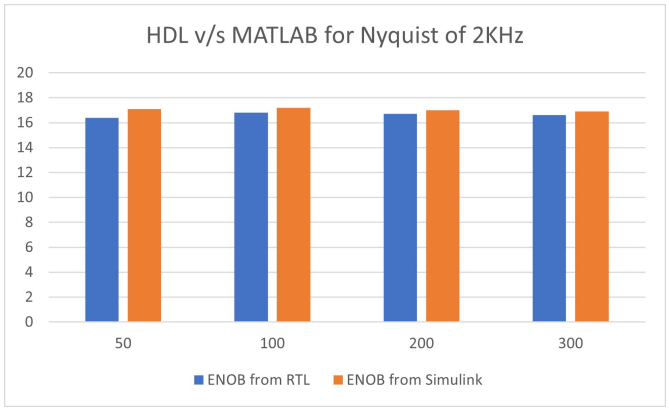
MODE	Parameter	HDL RESULTS
1	Nyquist = 1 kHz (ENOB, bits)	17.95
1	Nyquist = 1 kHz (SNDR, dB)	109.82
0	Nyquist = 2 kHz (ENOB, bits)	16.71
0	Nyquist = 2 kHz (SNDR, dB)	102.35

Simulation Results and Observations

RTL design and verification of the complete digital filtering chain was conducted using Xilinx Vivado. To generate the test stimulus, a sinusoidal input was simulated in a Sigma–Delta modulator within Simulink, and the resulting 1-bit PCM output was exported to a text file. This dataset was loaded into the testbench memory and read at the modulating frequency to drive the CIC, FIR, and HALF-BAND filters. The resulting simulated waveforms are presented in Fig. 21. All intermediate and final waveforms exhibit the expected staircase profile associated with discrete 20-bit digital samples. The final half-band output accurately tracks the sinusoidal envelope, validating the correctness of the applied scaling and truncation logic. The absence of saturation or overflow events further validates the selected word lengths. For reference, performance analysis metrics ENOB and SNDR for both modes at input signal frequency of 100 Hz are shown in Table 5, aligning with the expected trade-off at a fixed oversampling frequency of 4.096 MHz.



(a) Comparison for Nyquist Frequency of 1 kHz



(b) Comparison for Nyquist Frequency of 2 kHz

Figure 19: ENOB comparison between RTL simulation and MATLAB Simulink models across different signal frequencies.

Comparison with MATLAB results

For performing quantitative analysis, the 20-bit digital output samples generated by the RTL model were written to a text file through the testbench. This file was then imported into MATLAB, where ENOB and SNDR were computed for the RTL architecture and subsequently compared with the filter designs of SIMULINK. To validate the fixed-point RTL model against the floating-point Simulink reference, the Effective Number of Bits (ENOB) was computed at the filter output across various input signal frequencies for both the Simulink and HDL implementations. Since the RTL architecture supports Nyquist frequencies of 1 kHz and 2 kHz, comparative evaluations were carried out for both operating modes. The resulting performance metrics for these scenarios are presented in Fig. 19a and 19b.

VII. ASIC Flow

Synthesis Results (Timing, Area)

The synthesis stage translates the RTL description of the digital decimation filter into an optimized gate level netlist using the selected 180 nm CMOS standard cell library. During this process, the synthesizer performs logic optimization, resource sharing, and technology mapping while ensuring compliance with timing constraints. Key metrics obtained include timing performance in terms of Worst Negative Slack (WNS) and Total Negative Slack (TNS),total cell area and instance count as mentioned in Table[6],[7] . A positive WNS and zero TNS indicate that the design comfortably meets the required clock period. The synthesis results therefore provide the first quantitative validation of whether the architecture is feasible for ASIC implementation [17].

Table 6: Setup and Hold Time Summary

Setup Time Summary				**Hold Time Summary**			
Mode	**all**	**reg2reg**	**default**	**Mode**	**all**	**reg2reg**	**default**
WNS (ns)	4.664	9.349	4.664	WNS (ns)	0.141	0.141	1.374
TNS (ns)	0.000	0.000	0.000	TNS (ns)	0.000	0.000	0.000
Violating Paths	0	0	0	Violating Paths	0	0	0
All Paths	4543	2888	1680	All Paths	4543	2888	1680

Layout Summary (Utilization, Congestion)

Following synthesis, the placement and routing flow arranges all standard cell instances physically on silicon and establishes routing for interconnects as shown in Figure 20. The layout summary highlights essential parameters such as core utilization, routing density, and congestion levels across the die. For this design, a 2×2 mm core area was allocated, within which the synthesized digital filter components occupy approximately 0.4 mm² of active silicon. The congestion snapshot helps identify local hotspots where routing resources are heavily used, potentially affecting timing or manufacturability. Maintaining a balanced utilization and minimizing congestion ensures that the design can be cleanly routed without violating DRC (Design Rule Check) constraints (verified reports shared in zip folder). The final layout image confirm that the implemented architecture is physically realizable with acceptable density and routing margins.

Table 7: Cell Area and Instance Count Summary

Metric	**Value**
Total Cell Area (Cell + Physical) (μm^2)	415,576.837
Net Area (μm^2)	5,398.508
Total Area (Cell + Physical + Net) (μm^2)	420,975.345
Leaf Instance Count	8,145
Sequential Instance Count	1,668
Combinational Instance Count	6,477
Hierarchical Instance Count	65

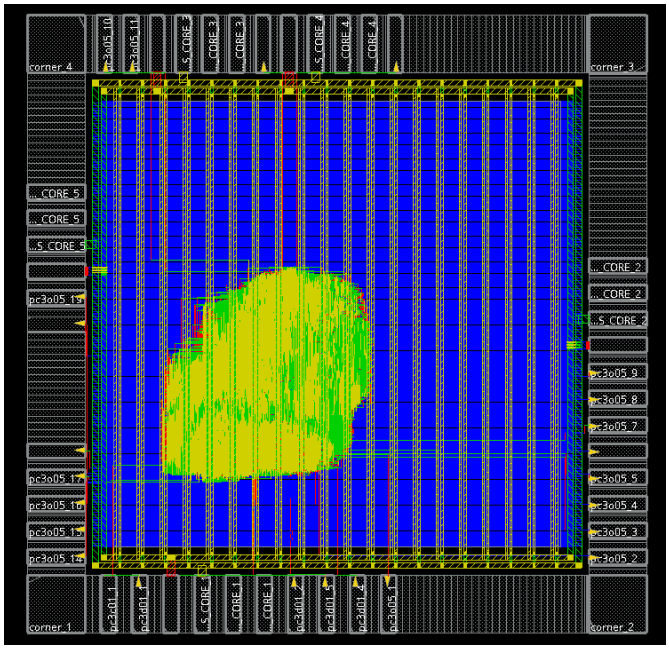


Figure 20: Placement and Routing View

Specifications vs Simulation Results

The implemented RTL architecture was evaluated against the key system-level specifications defined during the Delta-Sigma ADC design phase. The behavioral MATLAB/Simulink model provided the reference performance benchmarks, while the RTL simulations

and synthesis results were used to verify whether the digital backend remains consistent with these expectations. Functionally, the digital decimation chain exhibits output characteristics that align with the expected resolution targets of the modulator. The staircase quantized waveforms produced by the RTL when processed through MATLAB matches the intended ENOB and SNDR trends derived from MATLAB/Simulink model. Timing wise, the architecture was designed to support an operating frequency of 4.096 MHz, based on the decimation and sampling requirements. However, synthesis was intentionally performed at a slightly higher constraint (5 MHz) to introduce timing margin. The resulting positive slack obtained during synthesis indicates that the implementation comfortably supports the required clock rate with sufficient timing headroom across all critical paths. In terms of power, the synthesized design reports a leakage power of 14.6 microWatt, which is consistent with the expectations for a low power digital backend in mixed-signal ADC systems. This aligns with the use case of a Delta-Sigma ADC where the digital section must maintain high throughput without significantly impacting power efficiency. Overall, the RTL implementation behaves consistently with the design specifications, and its synthesis characteristics (timing, area, and estimated power) validate that the architecture is well within the operational limits defined during system-level modeling providing confidence that the physical implementation will uphold the intended ADC performance.

Post-Layout Simulation

The post-layout simulation was performed using the extracted netlist, and the waveform shown below reflects the behaviour of the design after placement and routing. While the internal nodes and clock activity appear as expected, the final output could not be observed in the post-layout simulation window due to some errors. Post layout environment was successfully generated and executed, the verification at this stage could not be fully consolidated. The waveform is therefore included in Fig. 21 as the obtained post-layout response for reference.

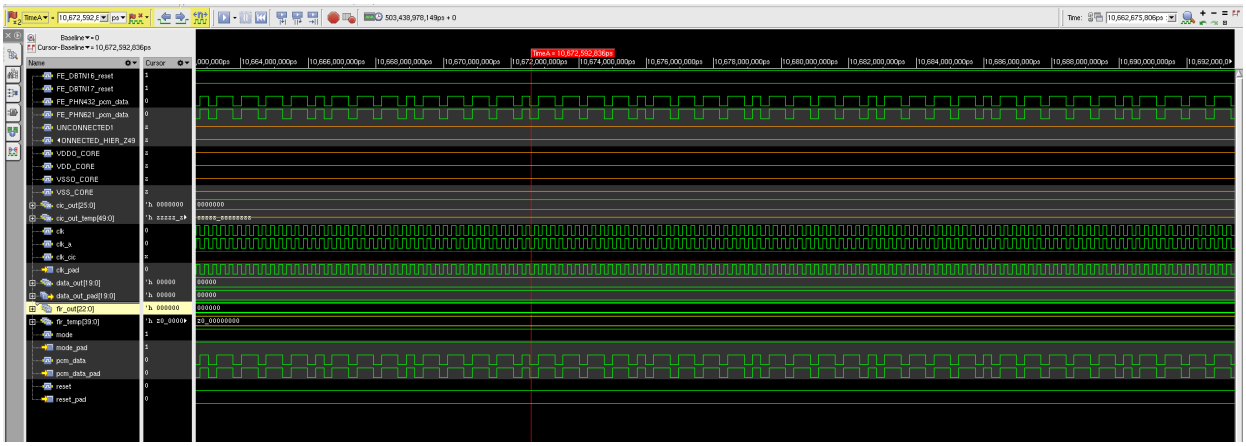


Figure 21: Post layout simulation waveform.

VIII. Challenges Faced

- **Long simulation windows for low-frequency noise resolution:** Since flicker noise dominates in the sub-kilohertz band, a large number of samples were necessary to achieve sufficient frequency resolution. This significantly increased Simulink simulation time and memory usage.
- **Matlab to HDL correlation:** Achieving bit-exact matching between the floating-point MATLAB model and the fixed-point Verilog implementation was challenging. Specifically, determining the optimal truncation points in the CIC and FIR filters to prevent overflow while minimizing quantization noise required multiple design iterations.
- **Navigating the Learning Curve of the ASIC Flow:** Transitioning from system-level modeling to physical implementation presented a steep learning curve. Mastering the complete RTL-to-GDSII flow required understanding complex tool-specific commands and constraints for Cadence Genus (synthesis) and Cadence Innovus (layout). Debugging set-up and hold time violations, along with understanding how logical constraints in the front-end directly impact physical congestion in the back-end, was a significant technical challenge for the team.

IX. New Learnings

- **Discovery of Hybrid CIFF-CIFB Architecture Benefits:** Through the comparative study of modulator topologies, we discovered that combining Feed-Forward (CIFF) and Feedback (CIFB) paths creates a hybrid architecture that outperforms standalone designs. We learned that this specific configuration allows us to minimize OTA output swing (saving power) via the CIFF path while retaining the superior stability and noise-shaping control of the CIFB loop, effectively solving the trade-off between power efficiency and robustness. (K_f, C_{ox}, W, L, and α) and observable system-level behaviour.
- **Practical construction of $1/f$ noise in time domain:** The work demonstrated how analytical PSD expressions for flicker noise can be transformed into realistic time-domain signals using inverse-FFT synthesis. This clarified the relationship between device-level noise parameters (K_f, C_{ox}, W, L, and α) and observable system-level behaviour.
- **RTL to GDSII Translation:** We gained practical exposure to the complete RTL to GDS flow, understanding how a functional RTL design is transformed step by step into a manufacturable GDSII layout. This included learning how synthesis maps RTL into standard cells, how placement and routing organize these cells physically, how timing and congestion are analyzed, we also learnt how DRC (Design Rule Check) ensures that the layout complies with all fabrication rules before tape-out.

X. Limitations & Risks

- **Unverified Impact of Clock Jitter:** While the report discusses the theoretical advantage of the Hybrid architecture in confining jitter sensitivity to the first stage, the current behavioral simulations utilized an ideal clock source. Quantitative analysis of clock phase noise and its direct degradation of the SNR was not performed. Consequently, the system's actual robustness against clock jitter in a real-world noisy environment remains unverified.
- **Truncation and Dynamic Range Boundaries:** As noted in the RTL design, the digital filter chain employs truncation to manage word-length growth (e.g., scaling the CIC output from 50 bits to 26 bits). While this logic is verified for standard operating ranges, input signals approaching the absolute theoretical limits (saturation boundaries) carry a risk of precision loss or unexpected quantization artifacts that could degrade the final ENOB.
- **Limited HDL Verification Coverage:** The RTL verification was primarily conducted using directed test vectors generated from MATLAB Simulink models. Comprehensive verification methodologies, such as Constrained-Random testing or Universal Verification Methodology (UVM), were not employed. Consequently, specific edge cases, illegal state transitions, or corner-case input sequences may remain untested, posing a risk that functional bugs could exist outside the standard sinusoidal operating conditions.

XI. Tools and Software Packages Used

- MathWorks' Matlab R2024a (Institute License), Windows 11
- Xilinx Vivado 2024.1 (Institute License), Windows 11
- Cadence Genus (v20.11-s111), Cent-OS
- Cadence Innovus (v20.14-s095), Cent-OS
- SCL 180 nm PDK (Process Design Kit)
- XCELIUM2009, Cent-OS

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